

MobilityStudio: Making MATSim Accessible

A Visual Workbench and a First Step Towards Conversational, AI-Assisted Transport Planning

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¹ MobilityStudio

² Senozon

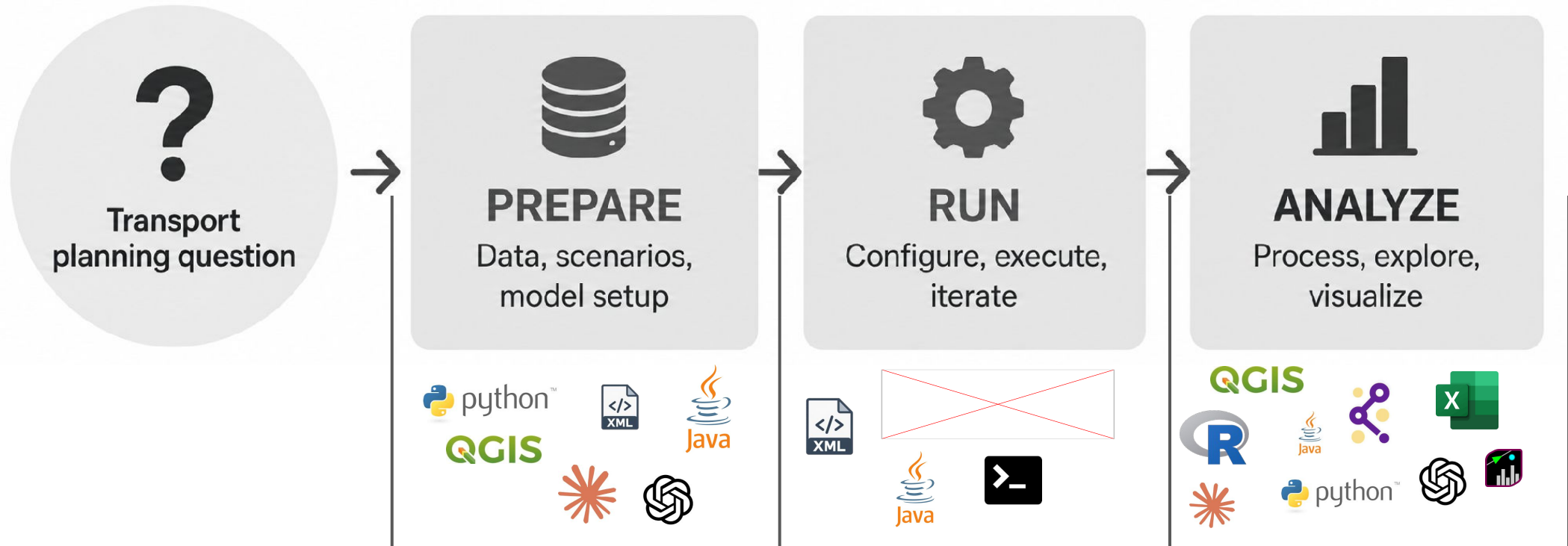


MobilityStudio

senozon

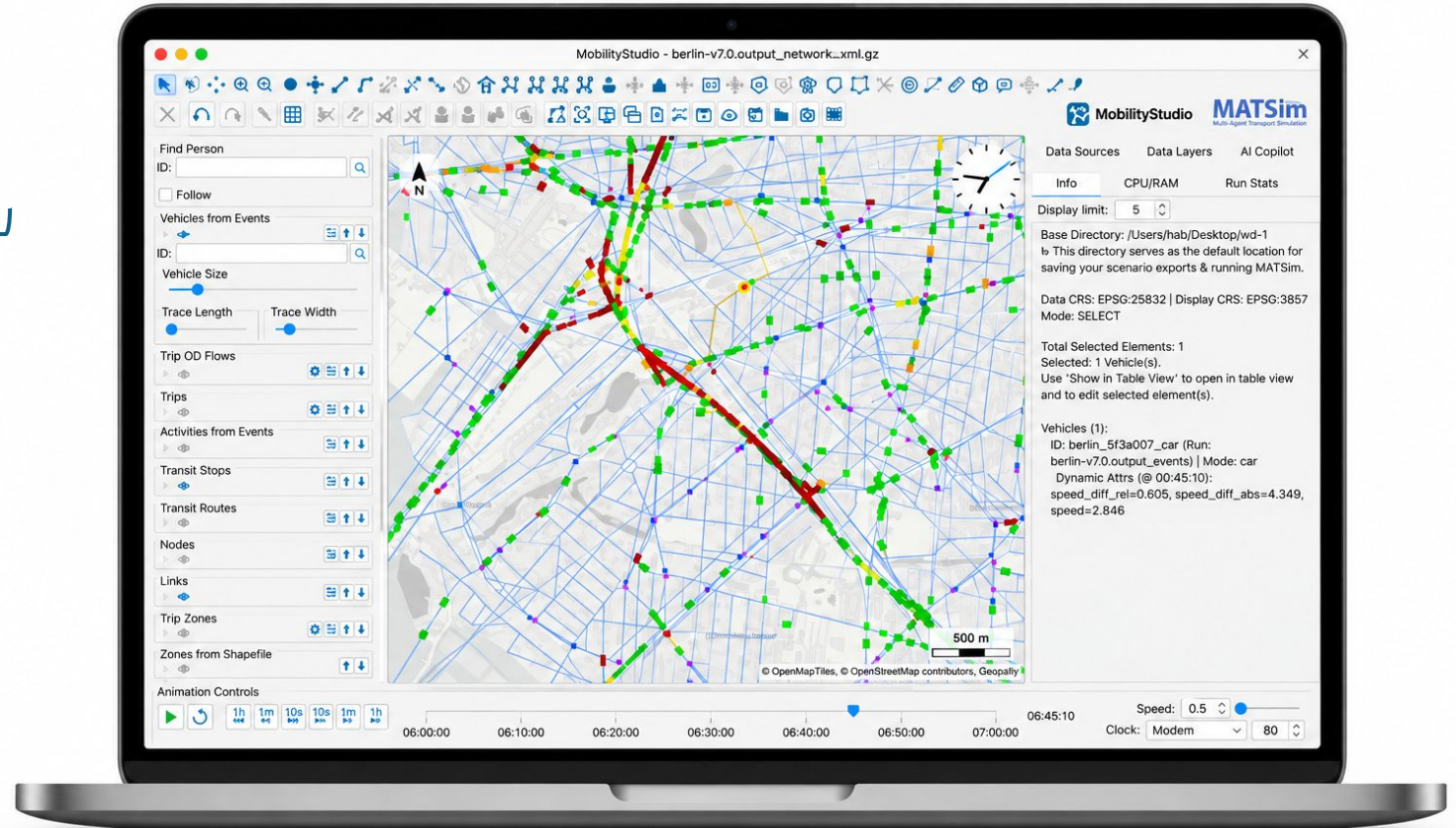
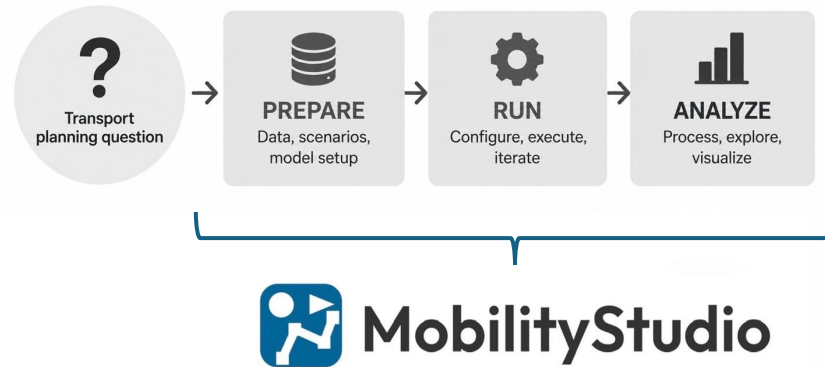
MATSim is powerful, but working with it can still be technical and complicated.

From a transport planning question to a model-based answer typically involves several steps and multiple tools.



 **Can we make this workflow easier?**

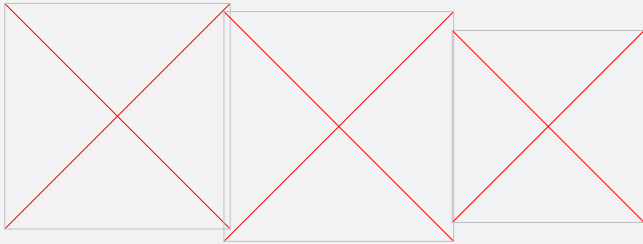
We built a visual workbench for MATSim.



Making MATSim accessible: Turning an expert simulation framework into intuitive transport planning software.

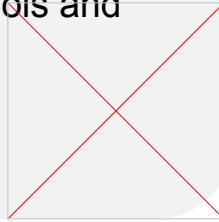
Core capabilities.

- A broad set of features, analysis and editing tools.
- Full support for the typical MATSim ecosystem: network, transit schedule, vehicles, facilities, population, counts, DRT, trips, legs, persons from CSV, events, ...



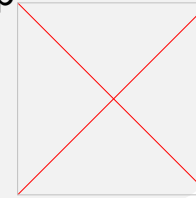
Entire workflow.

Create, run and analyze without (or with) switching between tools and files.



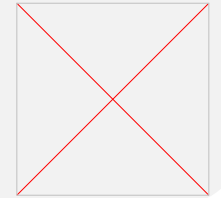
Map-first.

An intuitive GIS-like interface: Layers, selection tools, editing, map interaction.



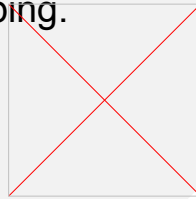
Map + Table.

Two synchronized ways to work with data.



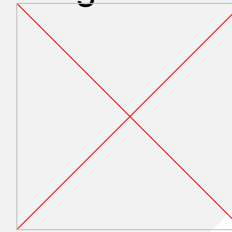
Static Assignment.

A fast path for static network assignment and prototyping.



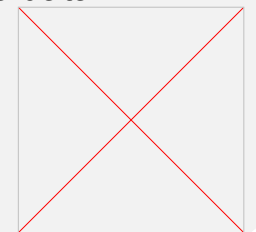
AI Assistance.

An integrated AI-Copilot in Chat mode and as Agent.



Demo tour.

The app starts with a guided demo tour and example data.





Activities from Plans

▼

ID:

Activity Size

Legs from Plans

>

Facilities

▼

ID:

Facility Size

☐ Draw Connectors

Transit Stops

▼

ID:

Stop Size

Transit Routes

>

Nodes

>

Links

▼

ID:

Link Width

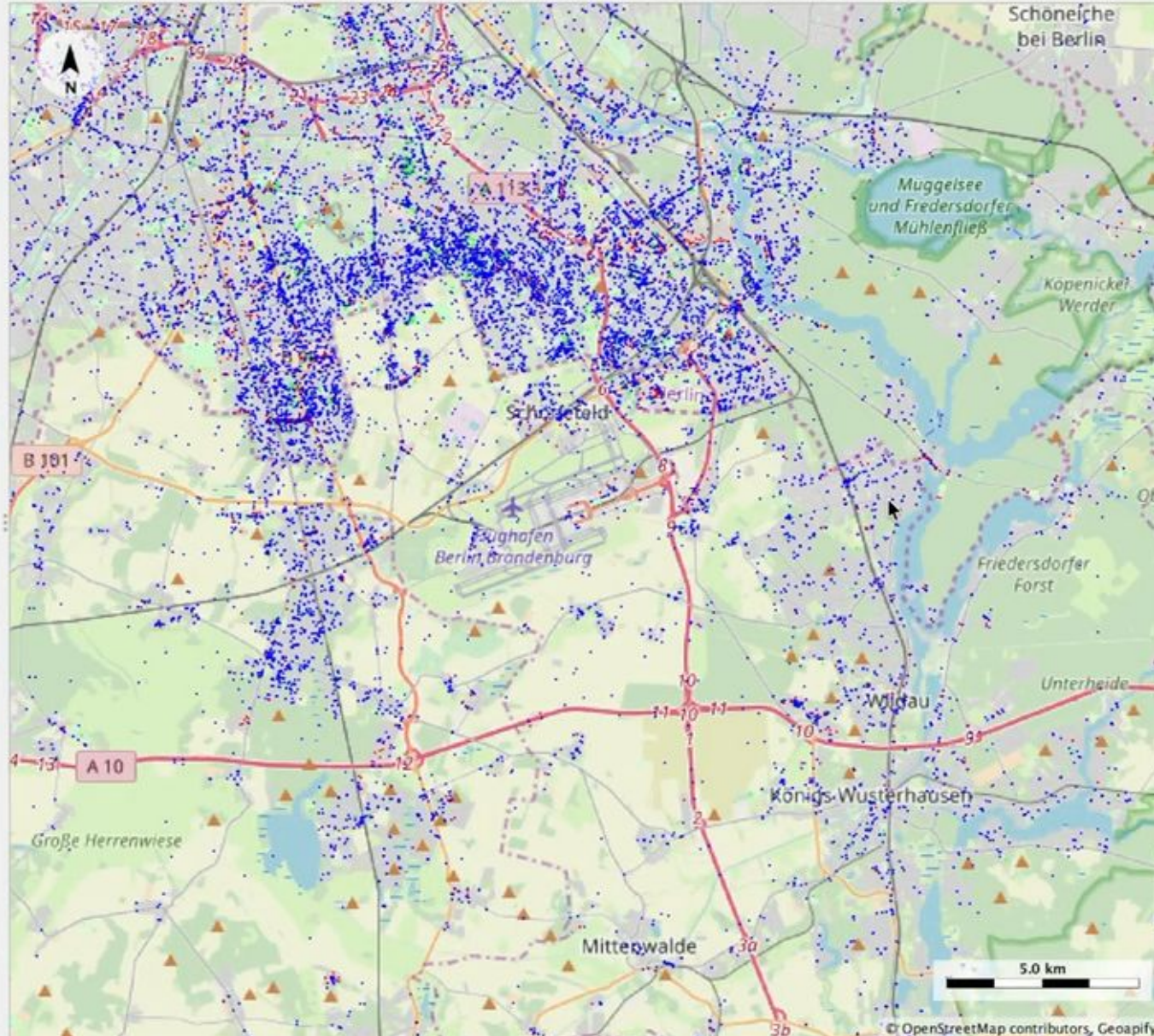
Link Offset

Background Map

> Style: OSM Standard ▼

Background Color

Choose Color



MobilityStudio



MATSim
Multi-Agent Transport Simulation

Data Sources

Data Layers

AI Copilot

Info

CPU/RAM

Run Stats

Display limit: 5 ▼

Base Directory: /Users/ihab/Desktop/wd-1

↳ This directory serves as the default location for saving your scenario exports & running MATSim.

Data CRS: EPSG:25832 | Display CRS:

EPSG:3857

Mode: MOVE_MAP

Click and drag to move the map.

Could we have asked the AI to do that?

The integrated AI Copilot in MobilityStudio...



Nodes

ID:

Node Size

Transit Stops

ID:

Stop Size

Transit Routes

Links

ID:

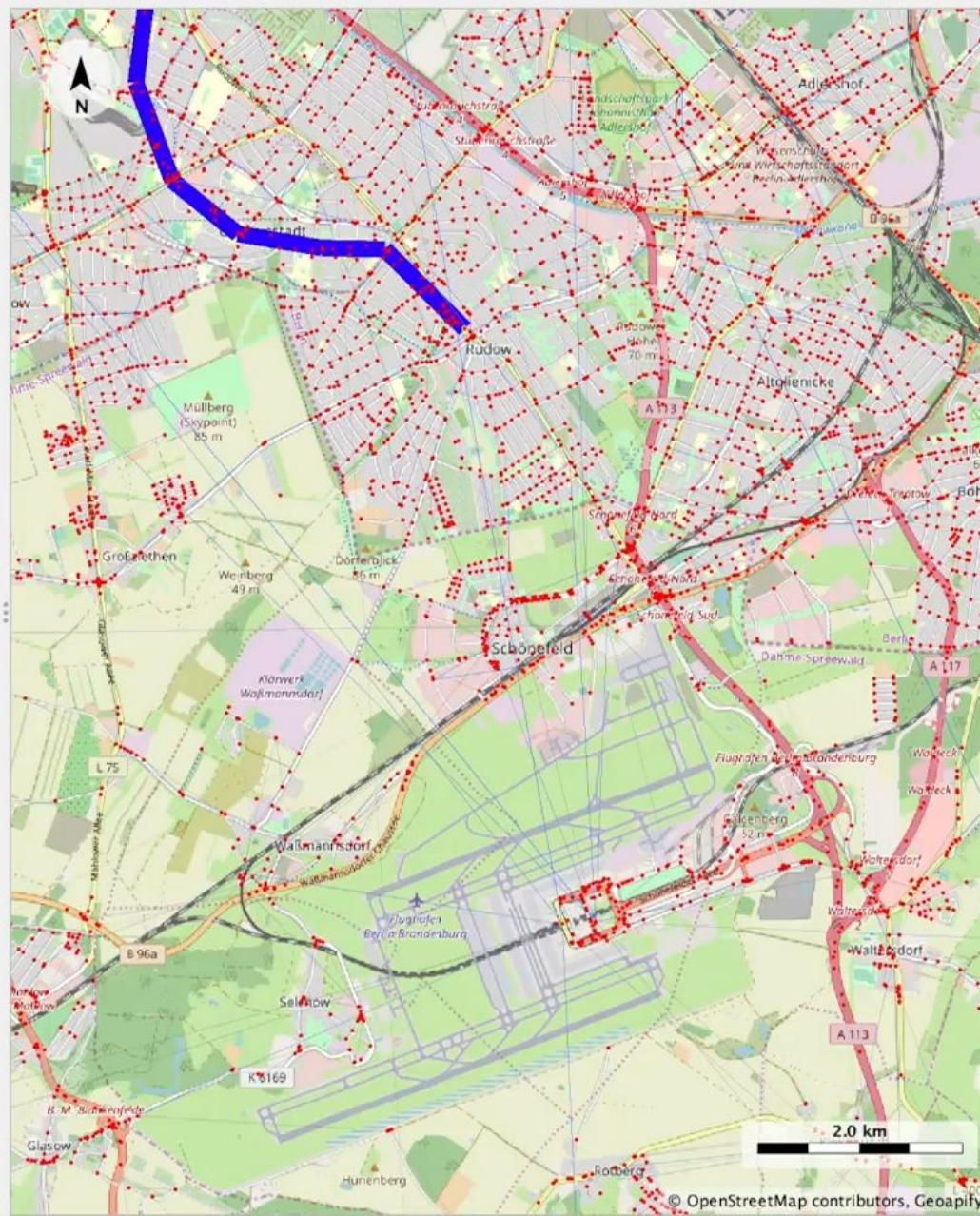
Link Width

Link Offset

Background Map

Style:

Background Color



Info CPU/RAM Run Stats Data Sources Data Layers AI Copilot

Settings Agent Mode ☒ Auto-approve agent edits

Welcome to MobilityStudio AI Copilot. 🌟

The AI Copilot has detailed knowledge about MobilityStudio in both Chat and Agent mode.

Use Chat Mode to ask about features and data. For example:

- "How do I add a new link?"
- "Where can I change link attributes?"
- "I have a link with attribute 'freespeed' and value 13.8889. What are the units of that value?"

Use Agent Mode to delegate tasks. For example:

- "Connect the two selected nodes by adding a road between them!"
- "Change the node color to green."
- "Implement a car restriction on the selected (yellow highlighted) links!"

Important: The Agent operates securely in an isolated zone ('ai_playground'). It can only read files from your working directory and cannot modify files outside the playground.

Expand Settings to configure your API key.

⚠ Note: The AI Copilot is experimental. We kindly recommend monitoring your token consumption and double-checking the results generated by the AI!

44 model(s) offered by Google Gemini.

New Session

View Context

Run Agent (Ctrl+Enter)

AI Usecase: Building scenarios

- We get what we get: a scanned overview map and a table from a planning report
- The problem: Planning documents don't speak MATSim
- Let's see what AI can do!*

1 Leerer Landstraße B72 Anbindung, Ortsteil Schirum

Nr.	Wohngebiet	WE	ha	Stand	Format
349	Stiegelhörner Weg	26	2,0	Rechtsverb.	pdf
Satz 63	Ortskern Schirum	22	18,4	Aufstellung	pdf

langfristige Siedlungsentwicklung

	Fläche	WE	ha	Stand
1	Fläche	10	1,1	SEK
2	Fläche	20	2,6	SEK

2 Kirchdorfer Straße K111 Anbindung, Ortsteil Kirchdorf

Nr.	Wohngebiet	WE	ha	Stand	Format
389	Weizenstraße	30	3,8	Auslegung	pdf

3 Oldersumer Straße L1 Anbindung, Ortsteil Rahe

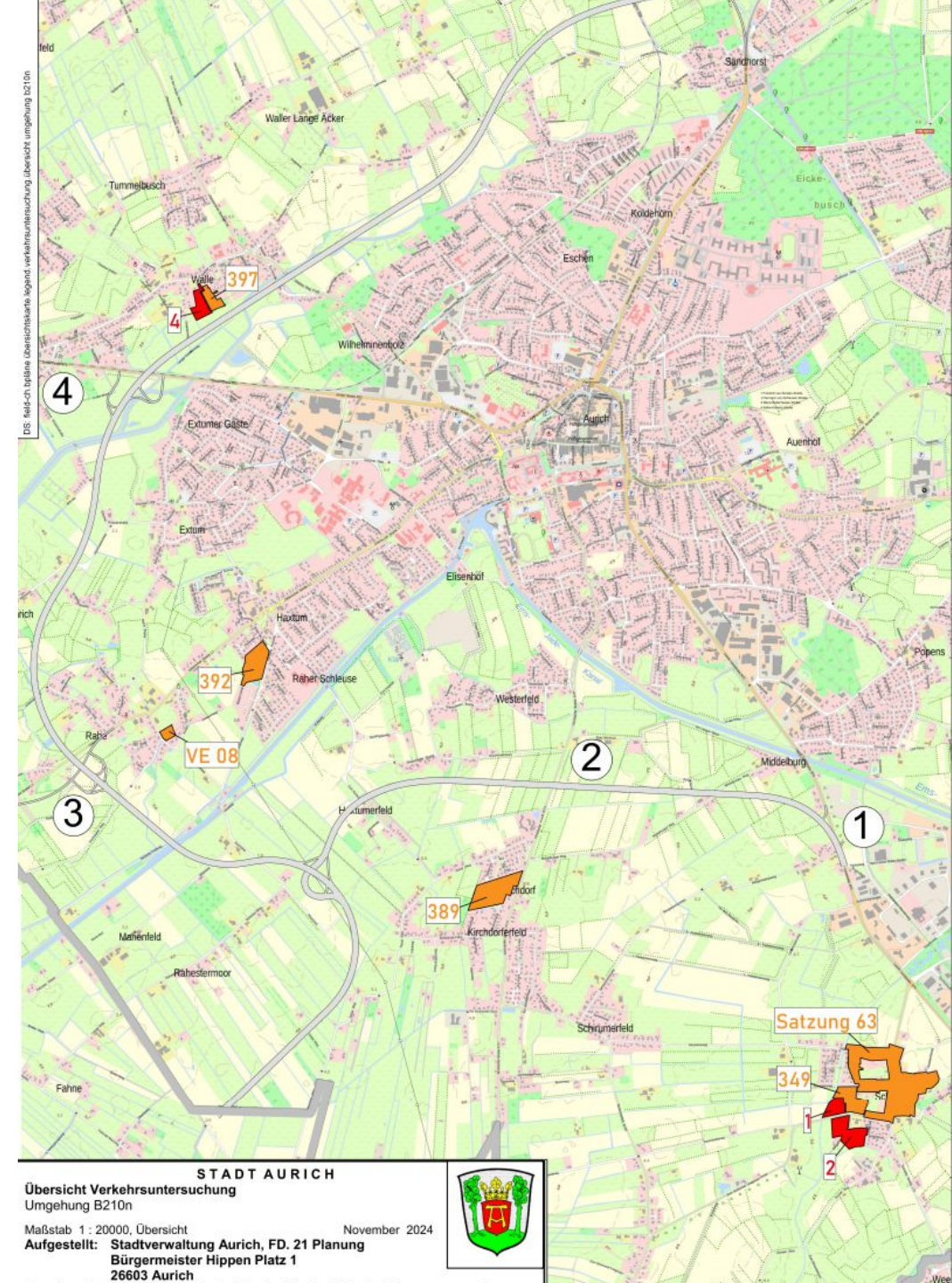
Nr.	Wohngebiet	WE	ha	Stand	Format
VE08	Rahe, Boomweg	20	5,5	Aufstellung	pdf
392	Hinter der Mühle	40	2,6	Aufstellung	pdf

4 Emden Straße B210 Anbindung, Ortsteil Walle

Nr.	Wohngebiet	WE	ha	Stand	Format
397	Südl. Binnen Diehl	12	1,1	Aufstellung	pdf

langfristige Siedlungsentwicklung

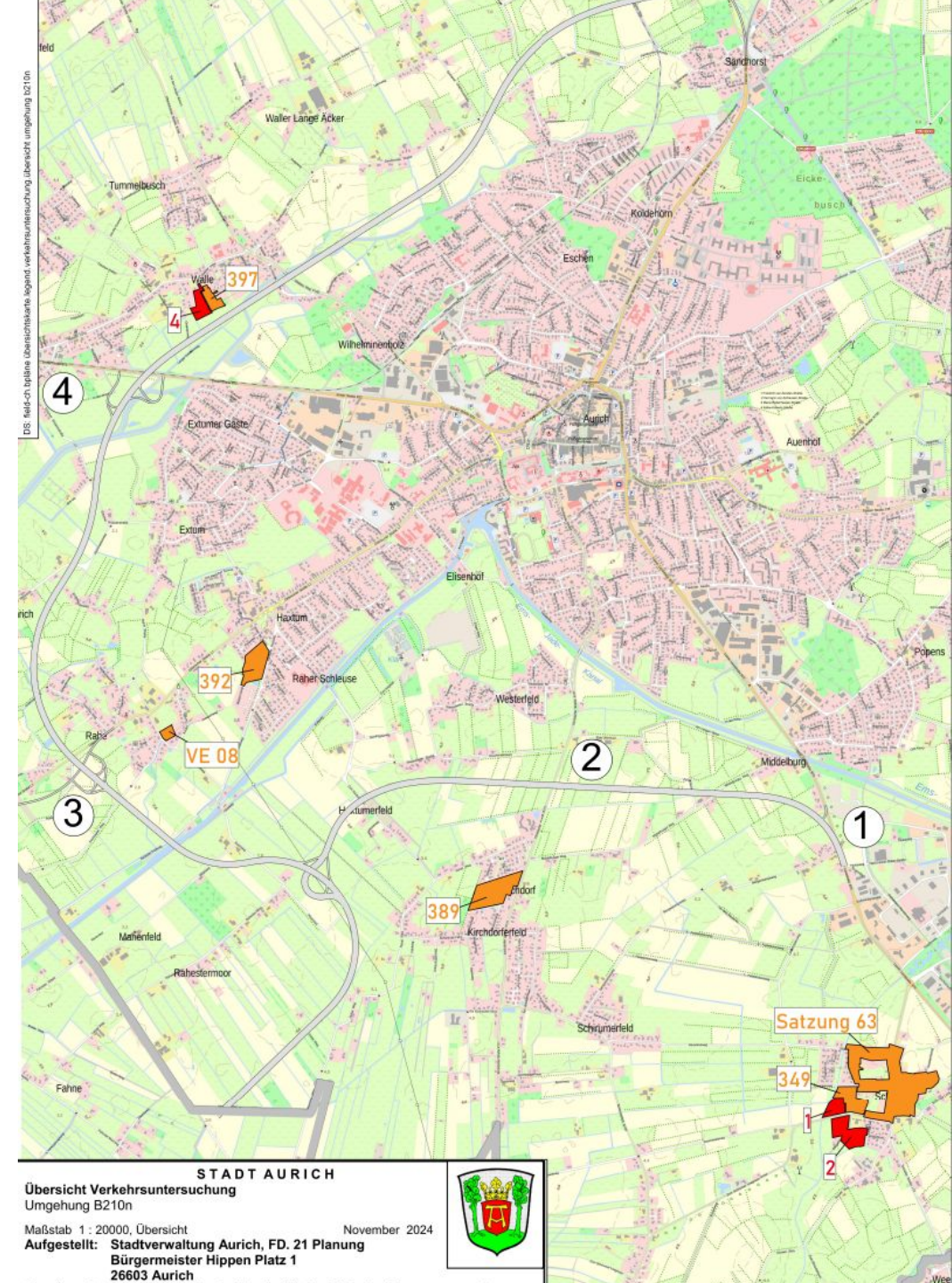
	Fläche	WE	ha	Stand
4	Fläche	10	1,3	SEK



*In the actual project this was done without AI

Building the Bypass B210n

“In this folder you see a MATSim network file and a pdf with a picture in it. The pdf depicts the area around the German town Aurich including a future road called B210n (in grey). The MATSim network file is of an area larger than the pdf file but fully includes it. The coordinates of the file are in EPSG:25832. By comparing the pdf picture to OpenStreetMap find out what the coordinates of the course of the B210n are as well as its connectors and insert it into the network file. Assume car,ride as the allowed modes and find a suitable freespeed yourself using other larger roads in the network as reference. If you find additional info about the B210n on the web you may use that too (all the info will probably be in German).”



Building the Bypass B210n

Georeference the image

Match landmarks and the printed scale to real-world coordinates (EPSG:25832) — the map has no embedded geodata of its own.

Isolate the B210n

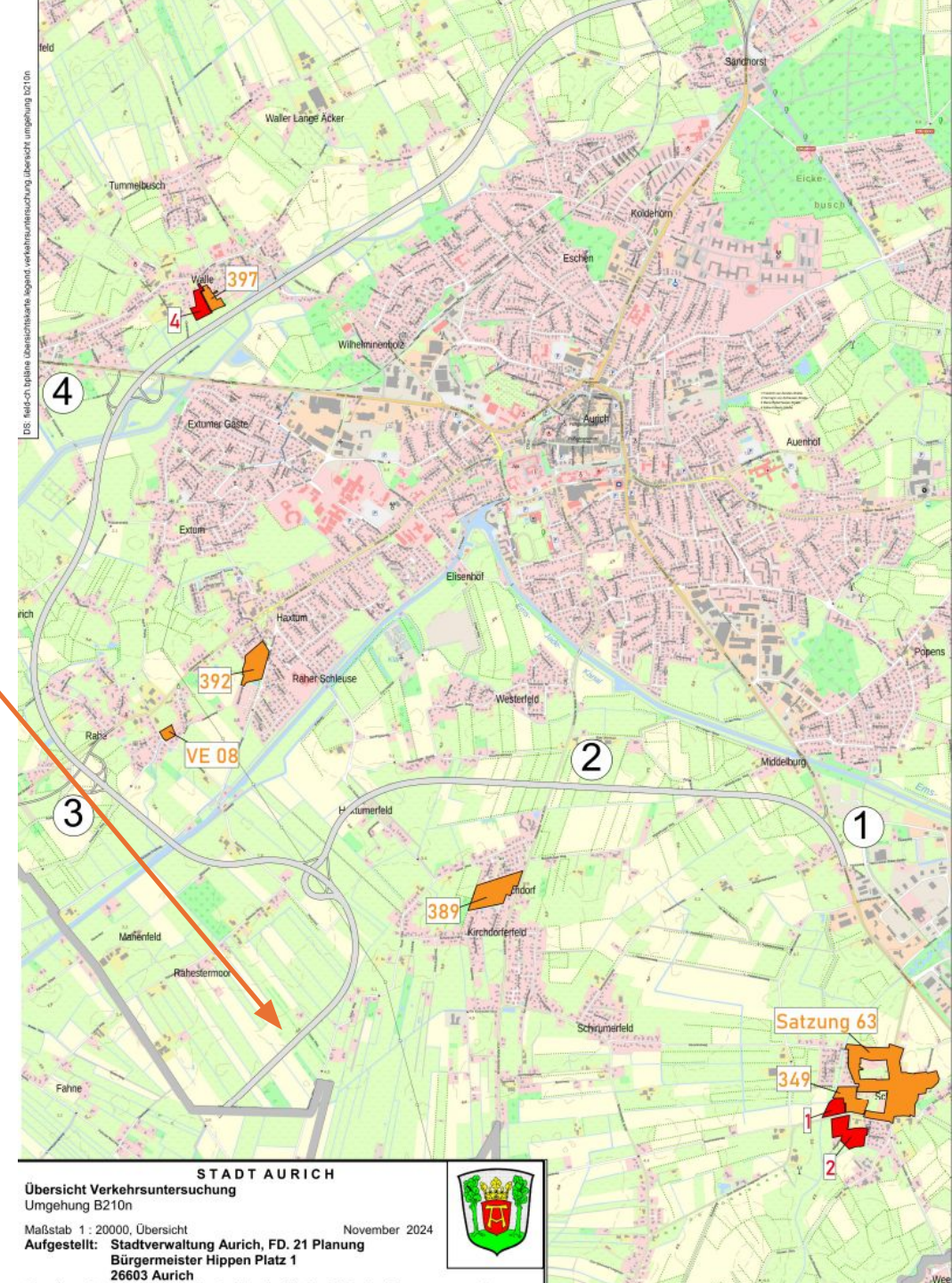
Figure out via online research that the south arm is not part of what we want to include, since it will be built in the very far future under a different name.

Choose realistic params

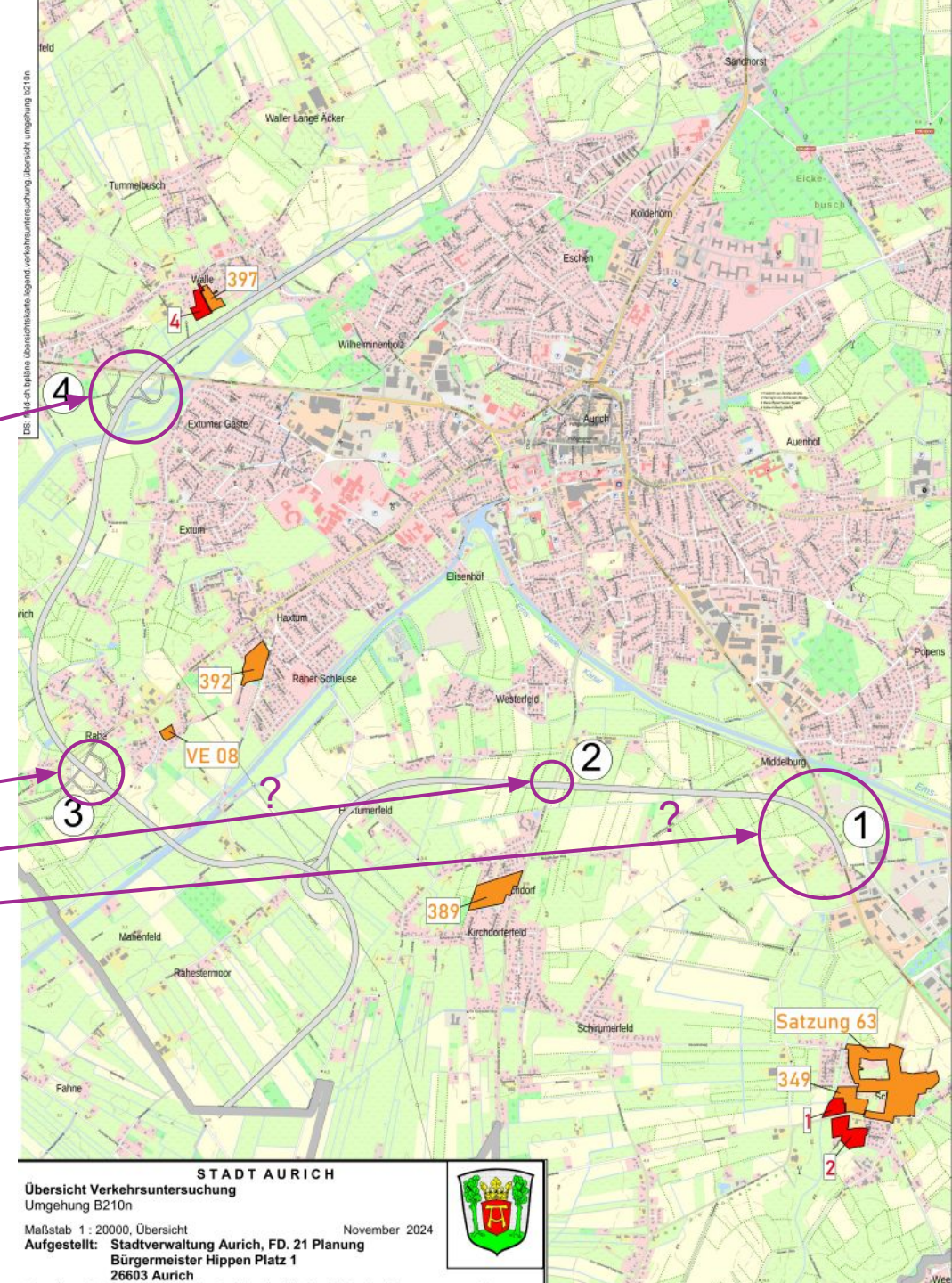
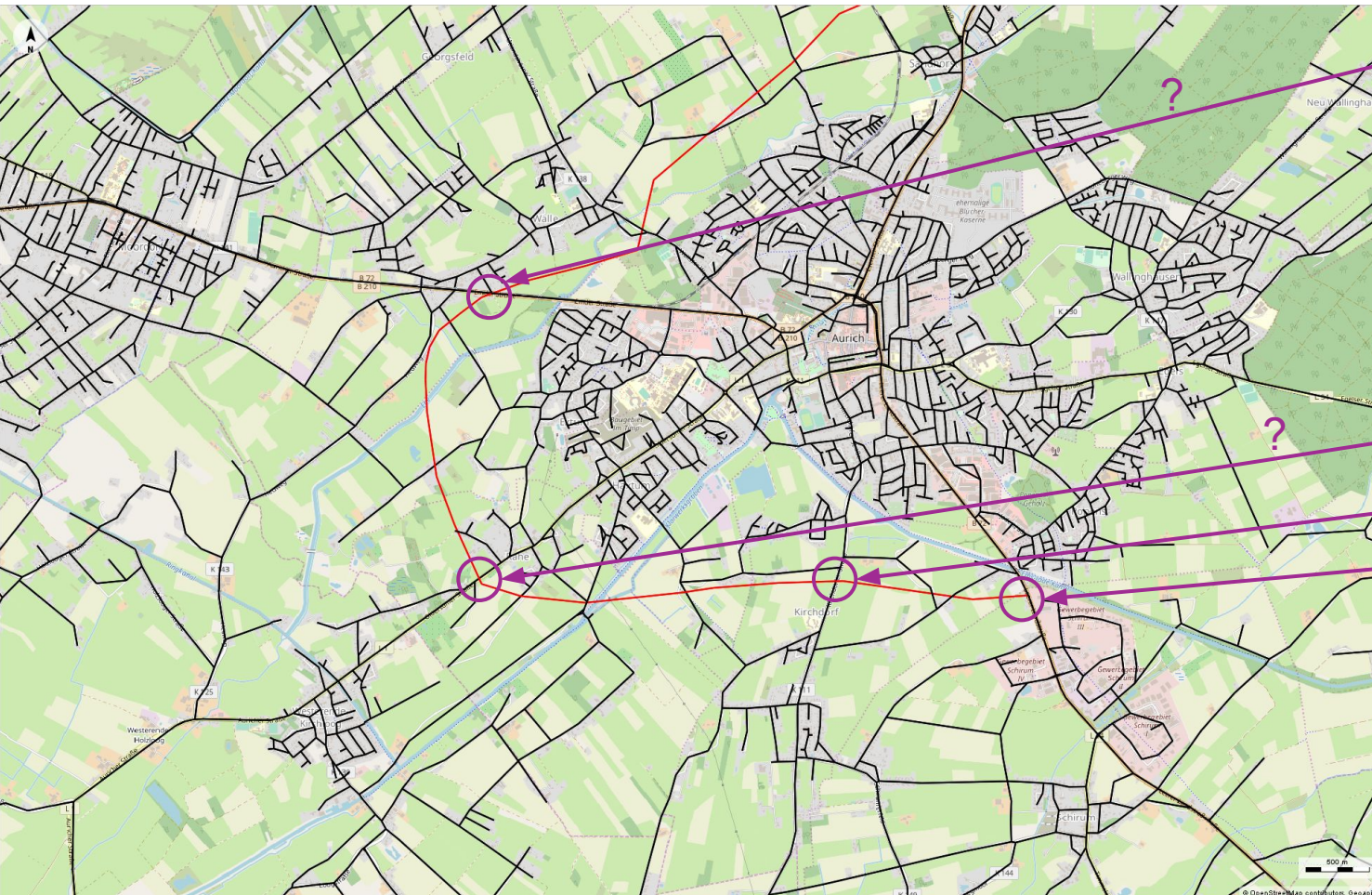
Set freespeed and capacity by comparison with similar classified roads already coded in the network — B-road vs. residential vs. ramp.

Trace the alignment

Follow the drawn route bend by bend, cross-checking every curve and junction against OpenStreetMap's existing road pattern for the same streets.



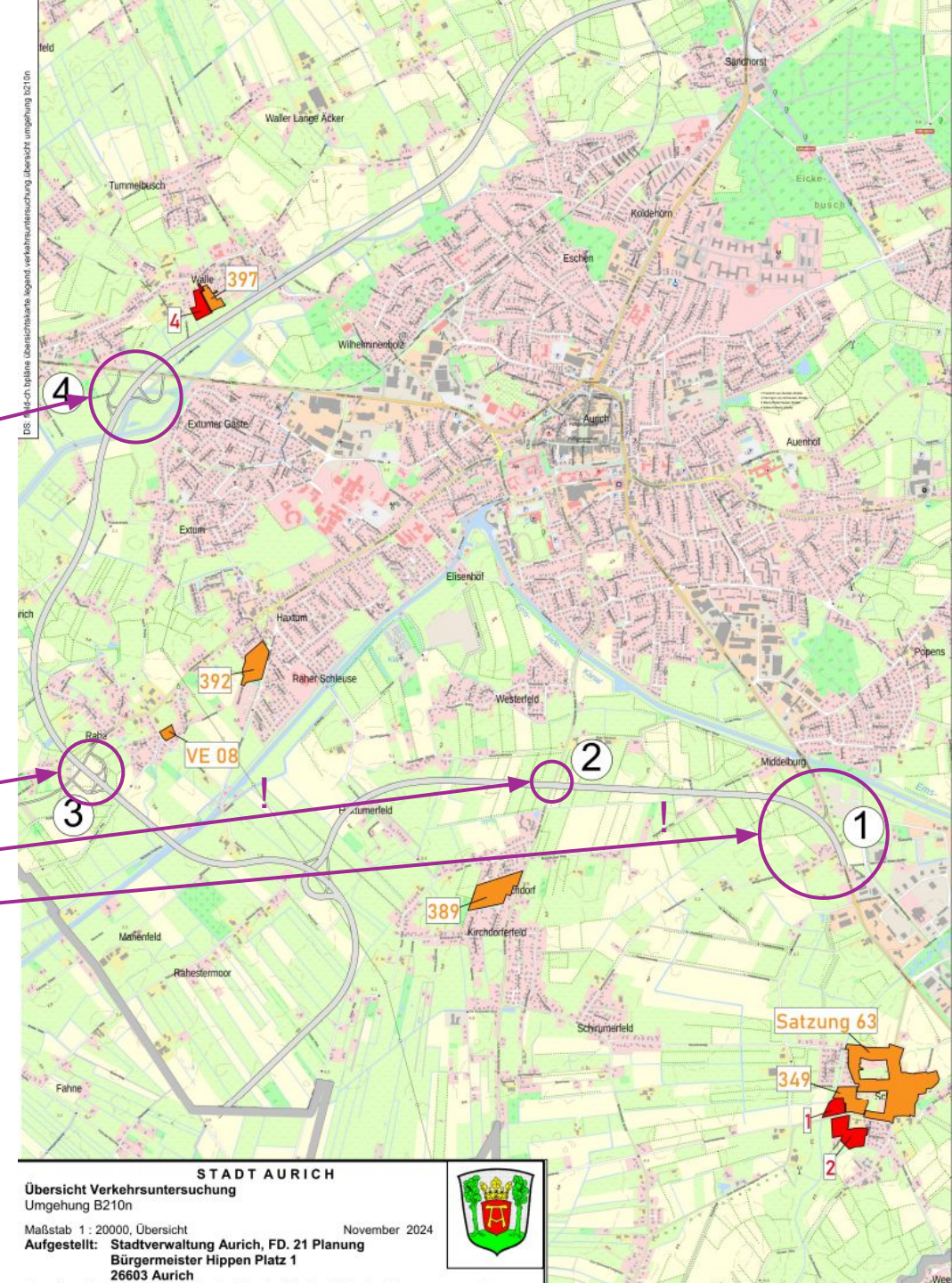
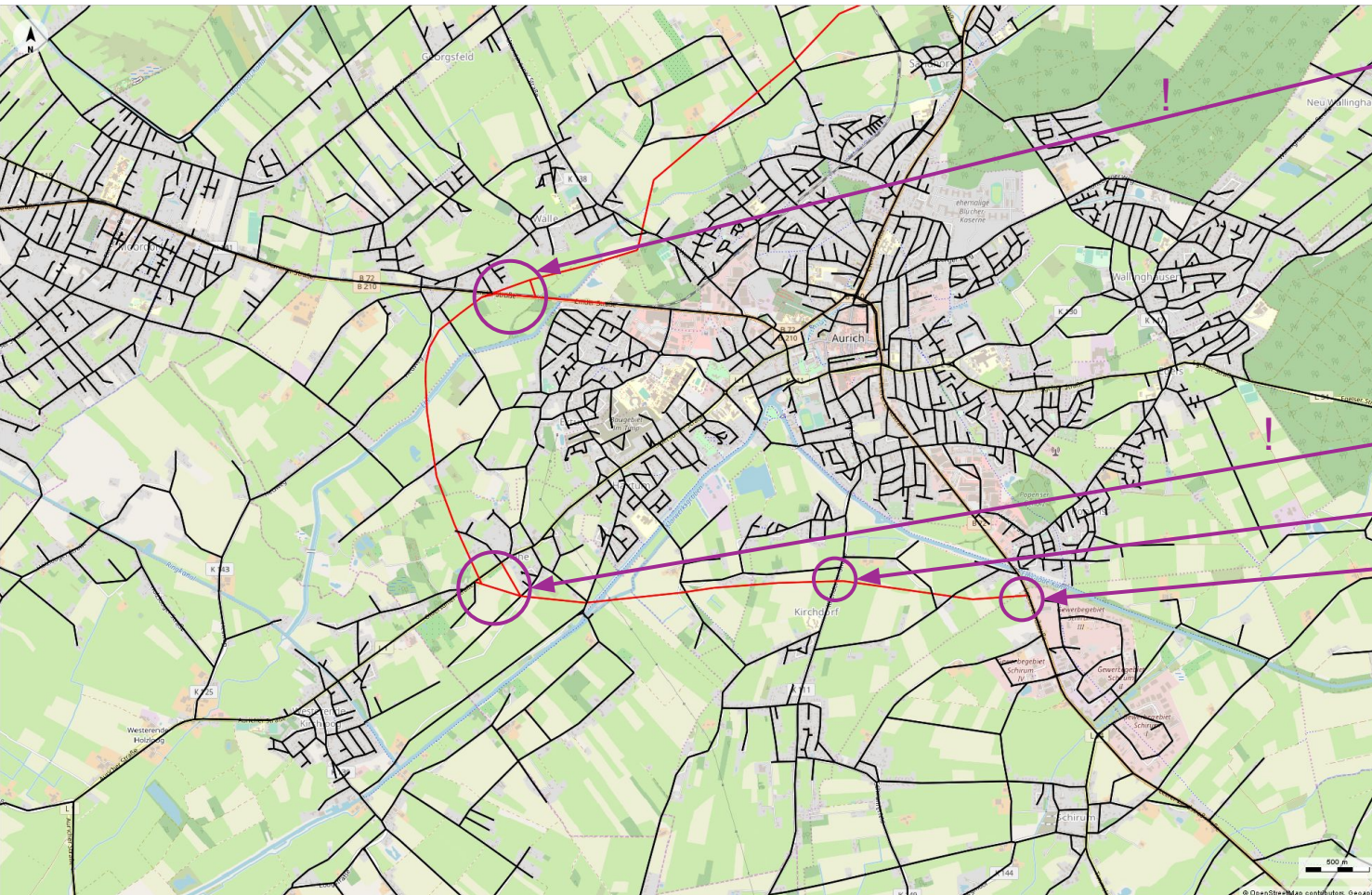
B210n – First Shot



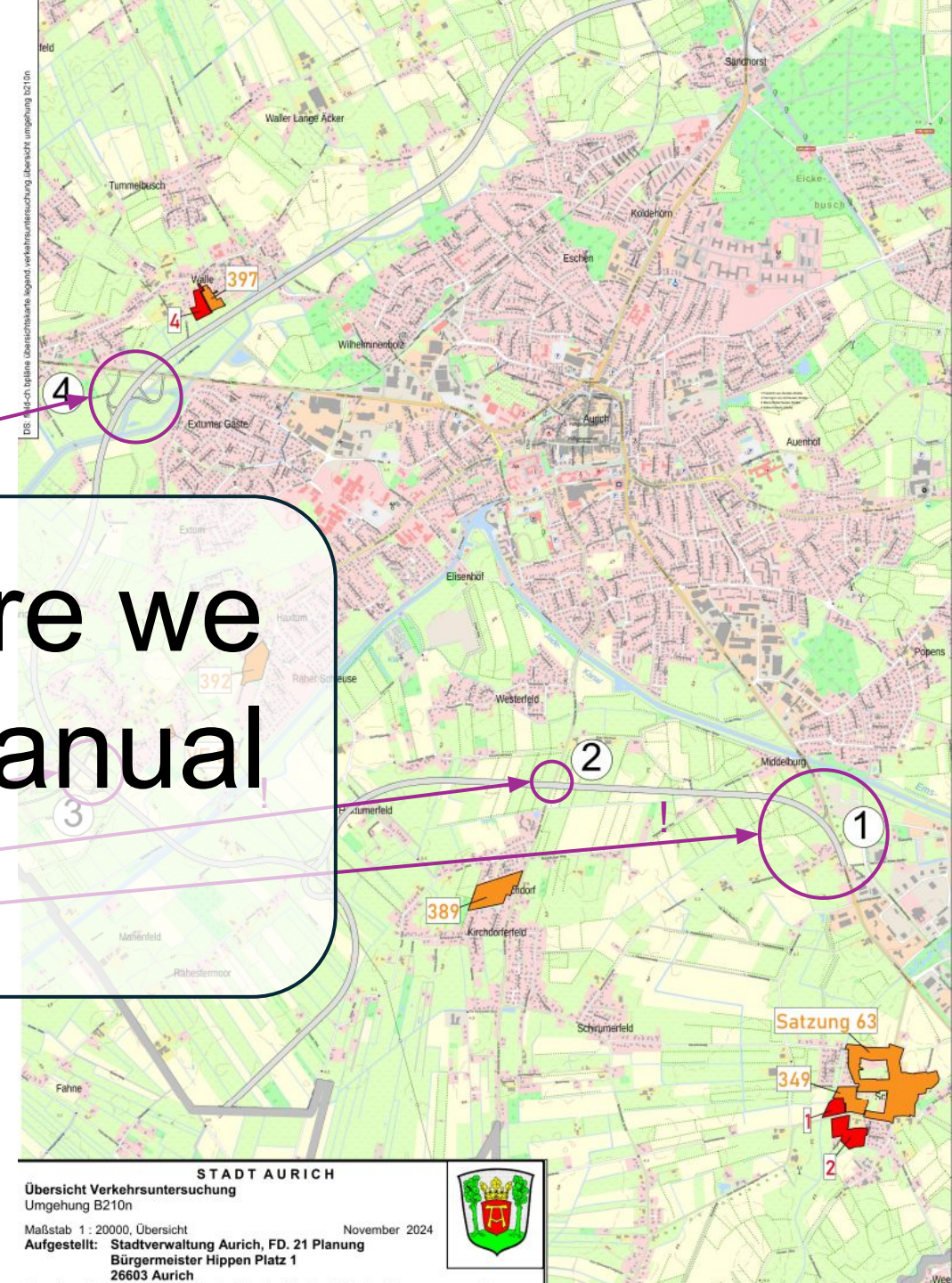
STADT AURICH
Übersicht Verkehrsuntersuchung
Umgehung B210n
Maßstab 1 : 20000, Übersicht
Aufgestellt: Stadtverwaltung Aurich, FD. 21 Planung
Bürgermeister Hippen Platz 1
26603 Aurich



After Some Iterations



After Some Iterations



New Home Facilities

Parse the pdf table

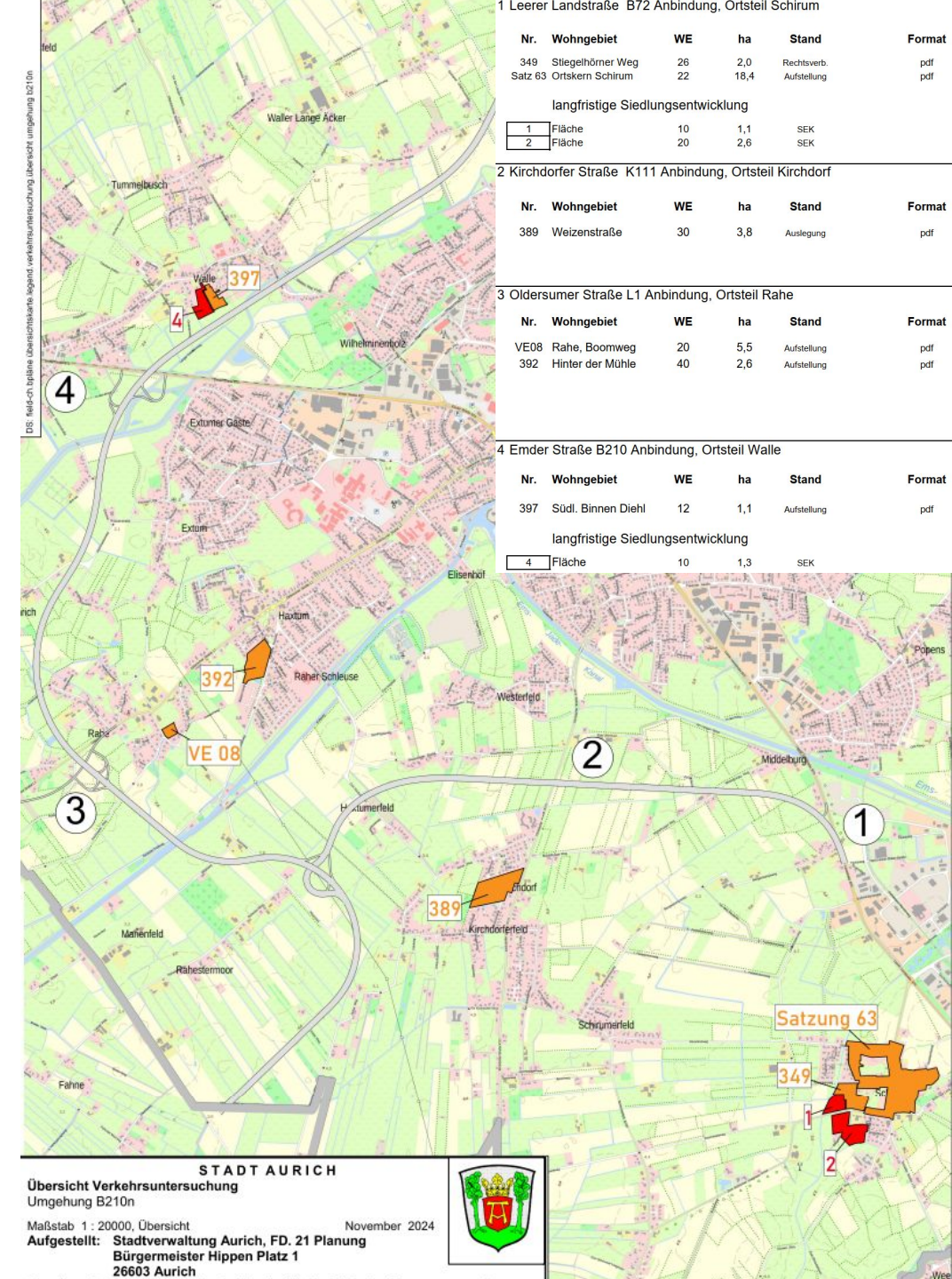
Find out how many flats (“WE”) will be located in each area.

Isolate the orange areas

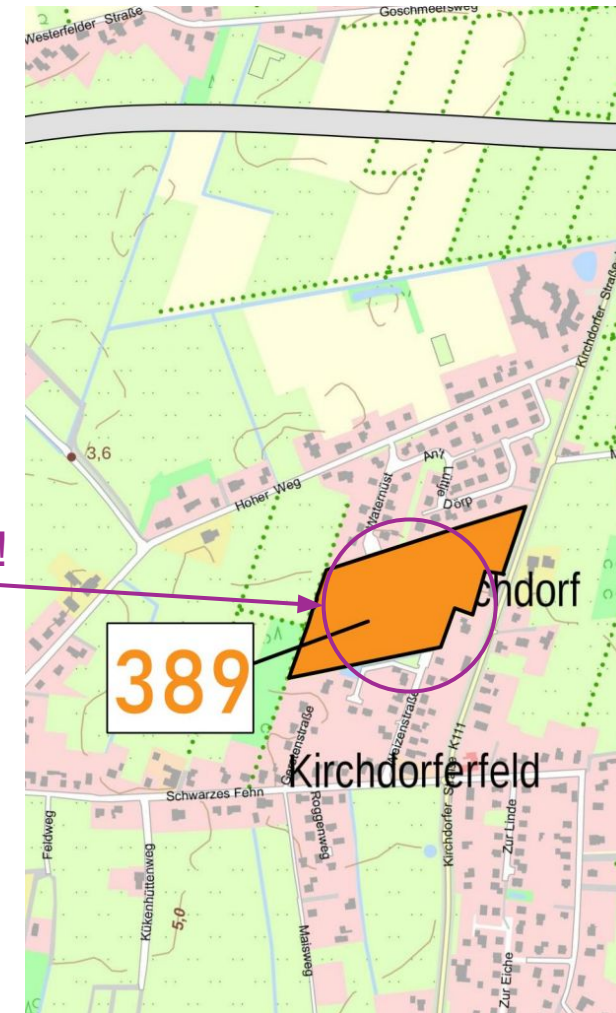
Geolocate the development areas and parse their labels and cross reference them with the table.

Place facilities

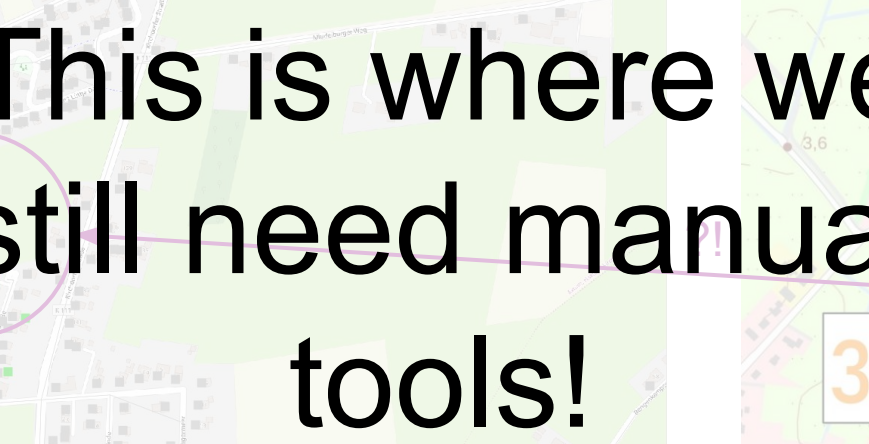
Place the correct number of facilities with reasonable capacities in the correct areas.



An aerial map of a residential area, likely in a rural or semi-rural setting. The map shows a network of roads, including a main road running vertically and several smaller roads branching off. A purple circle highlights a specific location on a road, with a purple arrow pointing to it from the right side of the map. The area is characterized by green fields, some buildings, and a few small structures. A scale bar in the bottom right corner indicates a distance of 100 meters. The map is credited to OpenStreetMap contributors and Geopapir.

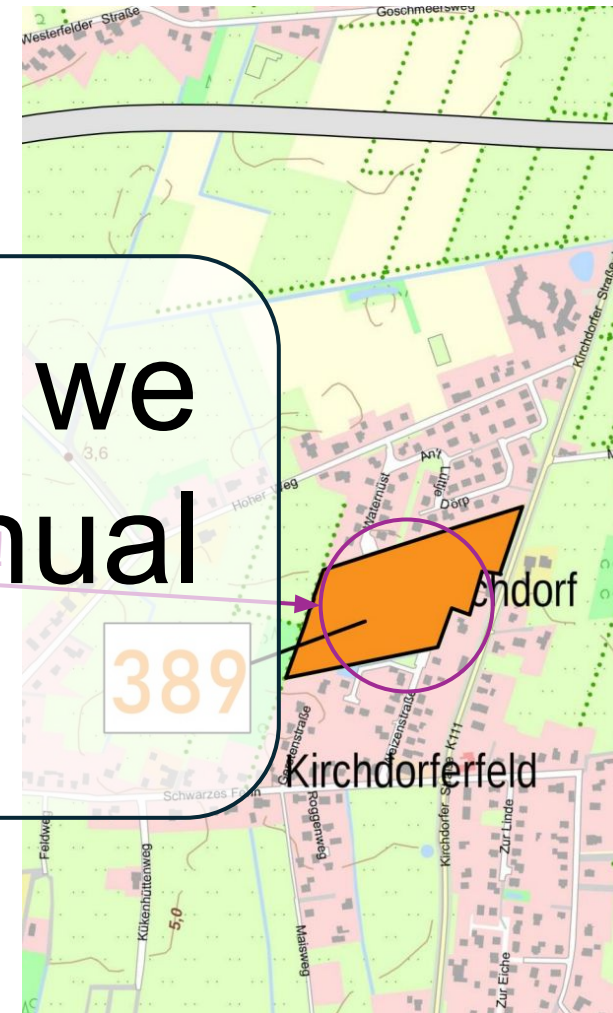


This is where
still need more
tools!

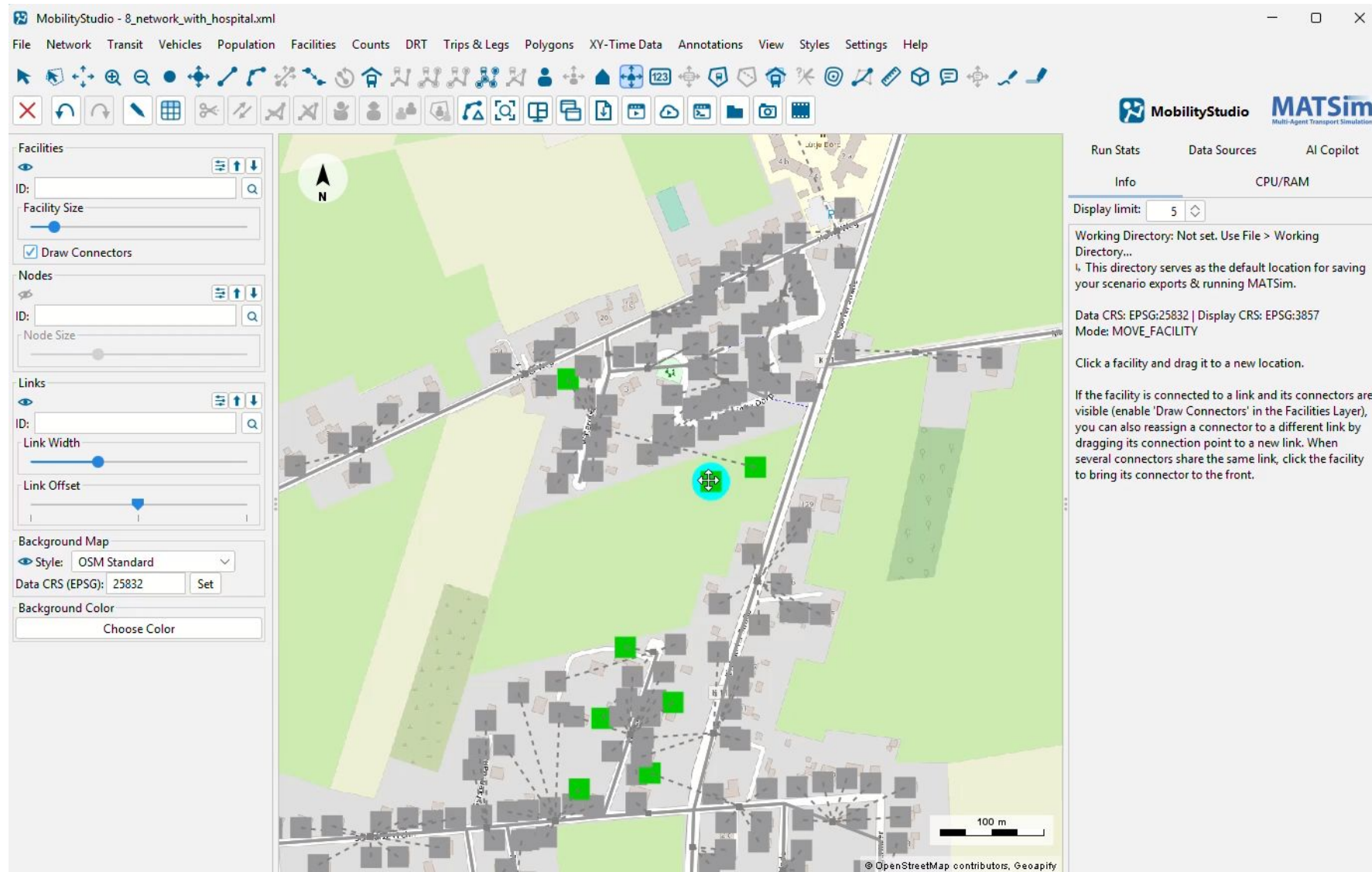


This is where we
still need manual
tools!

389



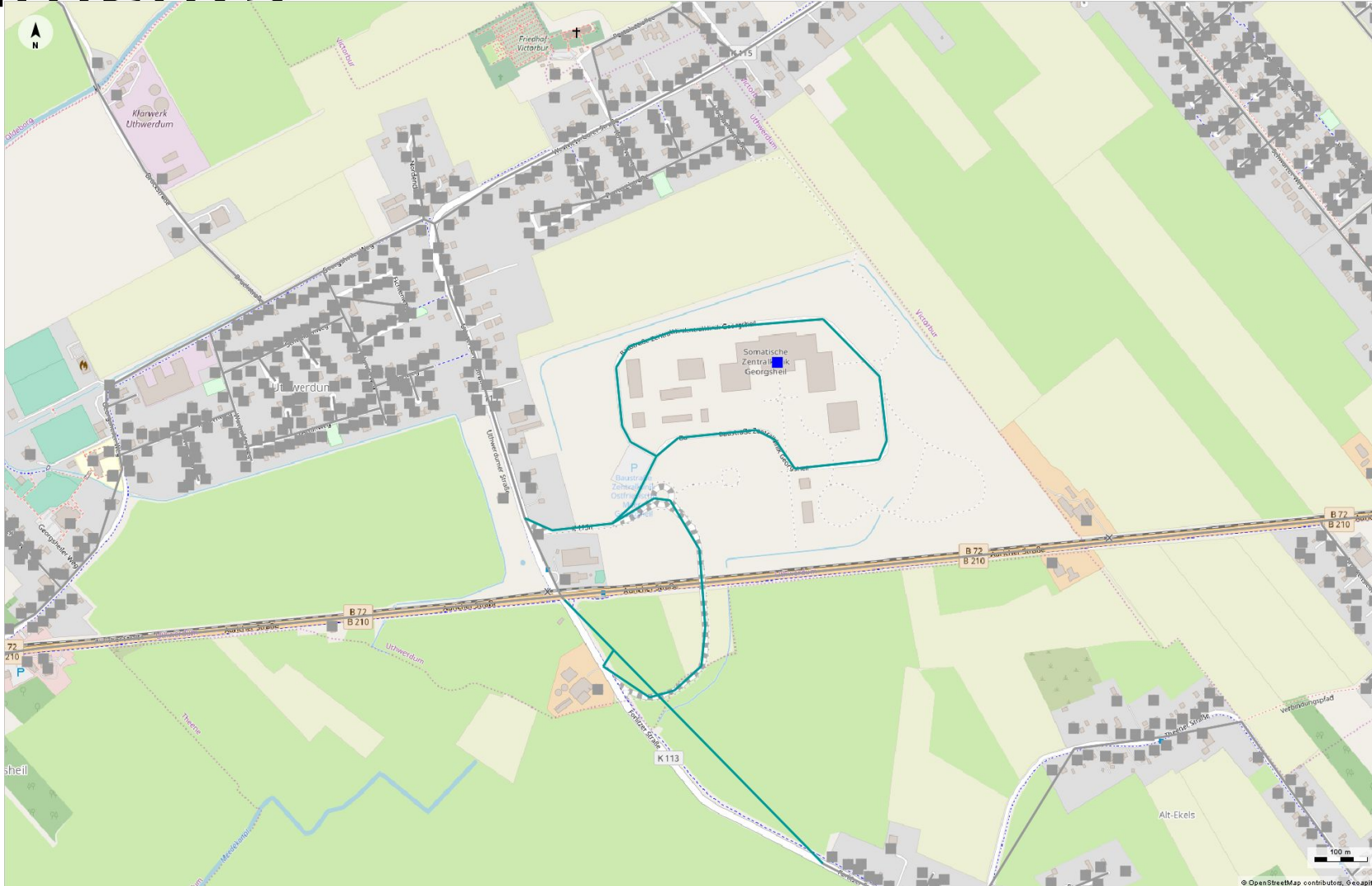
Fixing the Home Facilities



Final Test: Add something without documents

“do some online research about the "Zentralklinikum Georgsheil" (alternative name "Zentralklinik Ostfriesische Meere"). Place a facility where it is going to be. The facility should have [redacted] work capacity and an errand capacity of about 10% of all other errand capacities combined. Make new streets connecting the facility to the existing network. The planned layout of those streets is already in OpenStreetMap but you may also use other online sources. This is a complicated task. If you are unsure about something feel free to ask me.”

Final Test: Add something without documents



AI experiment takeaways written by AI:

TAKEAWAYS

What this changes — and what it doesn't

GOOD FIT FOR AI ASSISTANCE



Digitizing scanned plans into georeferenced network edits



Cross-referencing documents against a live map, at scale



Proportional / quantitative allocation across many zones



A reviewable first-pass extension, not a final answer

STILL NEEDS A DOMAIN EXPERT



Judging whether an interchange geometry is realistic



Knowing what a real junction or tunnel should look like



Deciding what belongs in scope for this scenario

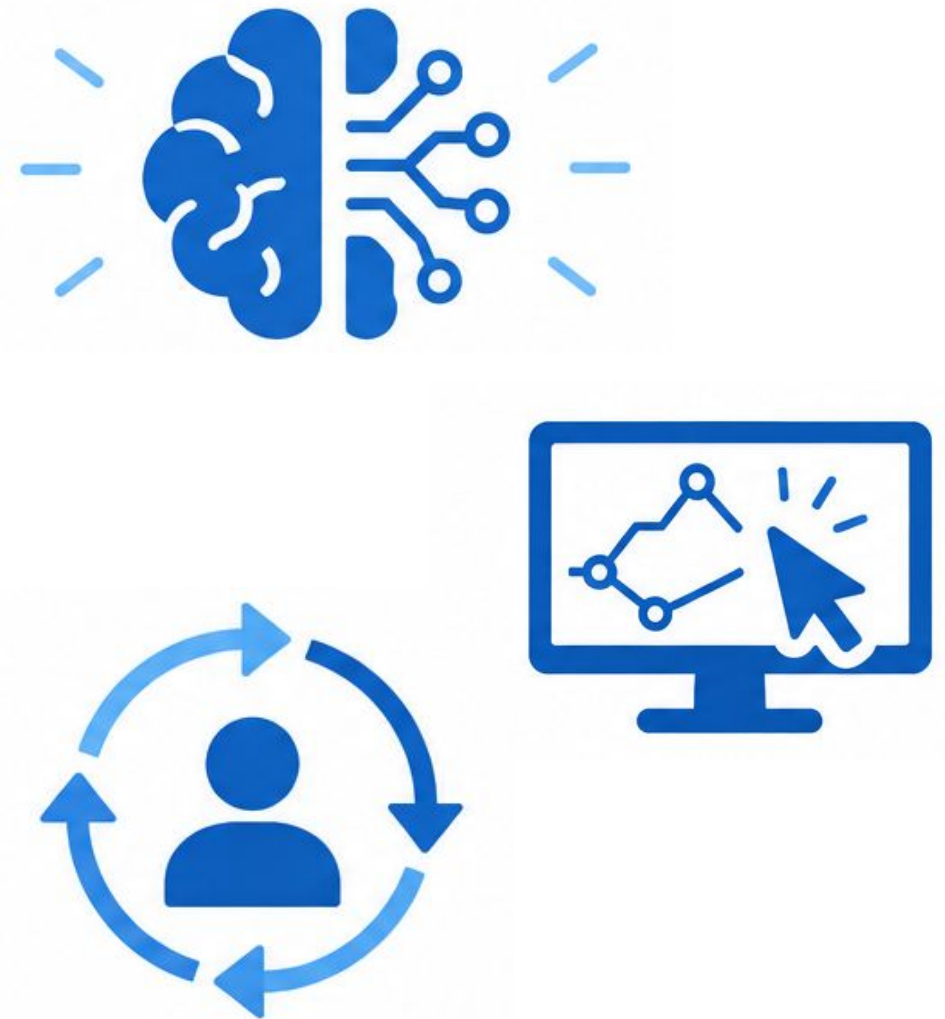


Final sign-off before the scenario is used for analysis

Every correction in this walkthrough came from a reviewer who knew exactly what to look for — that part doesn't change.

Conclusion

- MATSim is powerful (enough). Accessibility is the challenge...
- Much of the effort happens before and after the simulation. A graphical, integrated environment can improve existing MATSim workflows and lower the barrier for new users.
- AI is changing how we work with MATSim. But even with AI-assisted or agentic workflows, visual tools remain crucial:
 - to see, test and validate what AI did
 - to make precise edits intuitively (rather than telling the agent to do so)
 - to keep humans in the loop
- Both AI and intuitive tools like MobilityStudio can work together to make MATSim accessible to a much broader group of users.



Thank you!

**Want to download and try
MobilityStudio?**
www.mobilitystudio.de



**Need an application-ready
MATSim model?**
www.senozon.com

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